

Housing Reallocation and Demographic Transitions

A two-generation model

Environment

Agents. Time is discrete and one period is one generation. At date t , the economy contains Y_t young households and O_t old households. A young household has income y_i and liquid wealth b_i , drawn from a fixed type distribution F . A young renter becomes an old renter, a young owner becomes an old incumbent, and every old household exits at the end of the period.

Preferences. A young household values adult consumption c , space net of children's needs, and completed fertility:

$$u_t^Y(c, h, n) = \log c + \alpha \log(h - \kappa n) + \vartheta_t \log n. \quad (1)$$

Each child uses $\chi > 0$ units of goods and $\kappa > 0$ units of space. An old household values consumption, retained housing, and the estate delivered at death:

$$u^O(c^O, h^O, e) = \log c^O + \gamma \log h^O + \omega_B \log e. \quad (2)$$

All choices lie in the log domain: $c, n, h - \kappa n, c^O, h^O, e > 0$. A competitive financial sector trades a claim delivering one unit of next-period goods at price q , finances mortgages at the corresponding gross return, and buys stepped-up estate housing at the market price.

Housing and financial markets. A fixed stock $\bar{H}$ produces one unit of housing services per unit and period. Renters can occupy at most $\bar{h}^R$, while owners can occupy at most $\bar{h}^O > \bar{h}^R$. The asset price is P_t , one-period-ahead goods have price $q \in (0, 1)$, and $R = q^{-1}$. Competitive housing intermediaries imply the user-cost relation:

$$r_t = (1 + \tau^p)P_t - qP_{t+1}, \quad (3)$$

where $\tau^p P_t$ is the recurring property tax per unit.

Ownership and taxation. A buyer finances the share ϕ of the purchase with a one-period mortgage and posts $(1 - \phi)P_t h$ at closing. The mortgage advance is $\phi P_t h$ and its next-period repayment is $R\phi P_t h$. Selling a unit realizes the capital-gains tax

$$\ell_t = \tau^g [P_t - P_{t-1}]_+, \quad (4)$$

while a unit retained until death enters the estate with stepped-up basis. The tax is zero whenever the real asset price is constant. Only household sales are taxable events; stepped-up estate liquidations are exempt, and intermediaries trade at P_t with basis marked to market. Retained housing is occupied by the incumbent: households do not operate as landlords, and rental services are supplied only by intermediaries. The policy parameters τ^p and τ^g are constant. These cash flows give two tenure-specific costs. Buying a unit, paying its property tax, and selling it next period costs:

$$r_t^O = (1 + \tau^p)P_t - q(P_{t+1} - \ell_{t+1}) = r_t + q\ell_{t+1}. \quad (5)$$

For an old incumbent, retaining a unit rather than selling it, paying the property tax, and placing its next-period value in the estate costs:

$$r_t^I = (P_t - \ell_t) + \tau^p P_t - qP_{t+1} = r_t - \ell_t. \quad (6)$$

Thus the buyer anticipates the future tax, while the incumbent avoids the current tax by retaining the house.

Demography. Each child generates $\nu > 0$ future young households after survival and household formation. The children of a household affect future cohort size but do not inherit that household's individual state.

Household problems

Old households

An old incumbent enters with net financial wealth a and a house H . In current-value terms, selling the entire house produces $(P_t - \ell_t)H$; retained housing costs $r_t - \ell_t$ by equation (6). Her problem is therefore:

$$V_t^O(a, H) = \max_{c^O, h^O, e} u^O(c^O, h^O, e) \quad (7)$$

$$\text{subject to } c^O + qe + (r_t - \ell_t)h^O = a + (P_t - \ell_t)H + T_t, \quad 0 < h^O \leq H, \quad e \geq P_{t+1}h^O.$$

The estate includes the next-period value of retained housing. When its composition constraint and $h^O \leq H$ are slack and $r_t > \ell_t$, the incumbent's marginal value of retained space is:

$$\frac{u_h^O}{u_c^O} = \frac{\gamma c_t^O}{h_t^O} = r_t - \ell_t. \quad (8)$$

An old renter has no house and no realization. She maximizes $u^O(c^O, h^O, e)$ subject to $c^O + qe + r_t h^O = a + T_t$, $0 < h^O \leq \bar{h}^R$, and $e > 0$.

Young households

Conditional on renting, a young household chooses current consumption, housing, fertility, and saving:

$$W_t^R(i) = \max_{c,h,n,a'} u_t^Y(c,h,n) + \beta V_{t+1}^R(Ra') \quad (9)$$

subject to $c + \chi n + a' + r_t h = y_i + b_i + T_t, \quad 0 < h \leq \bar{h}^R, \quad a' \geq 0.$

Conditional on owning, the same household purchases h units, posts the down payment at the beginning of the period, and repays the mortgage when old:

$$W_t^O(i) = \max_{c,h,n,a'} u_t^Y(c,h,n) + \beta V_{t+1}^O(a_{t+1}, h) \quad (10)$$

subject to $c + \chi n + a' + [(1 - \phi) + \tau^P]P_t h = y_i + b_i + T_t,$
 $a_{t+1} = Ra' - R\phi P_t h, \quad (1 - \phi)P_t h \leq b_i,$
 $0 < h \leq \bar{h}^O, \quad a' \geq 0.$

Income received after closing can finance consumption and saving but cannot be pledged for the down payment. The household chooses the tenure with the higher value.

Suppose saving is interior and the future old retention limit is slack. For $m \in \{R, O\}$, let λ_t^m be the multiplier on the young budget, μ_t the multiplier on the owner's down-payment constraint, and η_t^m the multiplier on the tenure-specific size limit. The access values in current goods are:

$$\zeta_t^R = \frac{\eta_t^R}{\lambda_t^R}, \quad \zeta_t^O = \underbrace{\frac{\mu_t}{\lambda_t^O}(1 - \phi)P_t}_{\zeta_t^{O,F}} + \underbrace{\frac{\eta_t^O}{\lambda_t^O}}_{\zeta_t^{O,S}}. \quad (11)$$

Using the saving Euler equation and the old value-function envelopes, the housing conditions reduce to:

$$\frac{\alpha c_t^m}{h_t^m - \kappa n_t^m} = r_t^m + \zeta_t^m, \quad r_t^R = r_t, \quad r_t^O = r_t + q\ell_{t+1}. \quad (12)$$

The ownership expression follows from the purchase, mortgage, property-tax, and resale cash flows in equation (10). Equivalently, the access value is the household's marginal valuation above its tenure-specific user cost:

$$\zeta_t^m = \frac{\alpha c_t^m}{h_t^m - \kappa n_t^m} - r_t^m. \quad (13)$$

Thus the wedge is expressible from the household's bundle and prices, not only from multipliers.

The fertility condition prices the goods and space required by a marginal child:

$$\frac{\vartheta_t c_t^m}{n_t^m} = \chi + \kappa(r_t^m + \zeta_t^m). \quad (14)$$

A housing constraint raises the household's shadow price of a child by $\kappa\zeta_t^m$.

Steady states and transition

Let G_t denote the normalized distribution of old net financial wealth, inherited housing, and tenure. It is the pushforward of F under the preceding young policies:

$$G_{t+1} = \text{Law}_{i \sim F}(Ra'_t(i) - \mathbf{1}\{m_t(i) = O\}R\phi P_t h_t(i), \mathbf{1}\{m_t(i) = O\}h_t(i), m_t(i)). \quad (15)$$

Let $\bar{n}_t$, $\bar{h}_t^Y$, and $\bar{h}_t^O$ denote fertility and housing averaged over F and G_t , respectively. Let $\bar{s}_t^O$ denote housing sold by the average old household, assigning zero sales to old renters. Cohort sizes and the housing stock obey:

$$Y_{t+1} = \nu \bar{n}_t Y_t, \quad O_{t+1} = Y_t, \quad Y_t \bar{h}_t^Y + O_t \bar{h}_t^O = \bar{H}. \quad (16)$$

Fertility determines the next young cohort, while current young and old households compete for the fixed housing stock.

The government rebates property- and capital-gains-tax receipts equally across living households:

$$(Y_t + O_t)T_t = \tau^P P_t \bar{H} + \ell_t O_t \bar{s}_t^O. \quad (17)$$

The taxes remain private wedges, while their revenue stays within the household sector.

For a constant fertility preference ϑ , let $\mathcal{N}(P, T; \vartheta)$, $\mathcal{H}^Y(P, T; \vartheta)$, and $\mathcal{H}^O(P, T; \vartheta)$ denote stationary fertility and housing choices aggregated across household types.

Definition 1 (Positive steady state). A positive steady state is a constant equilibrium with $Y^* = O^* > 0$, a constant asset price and transfer, and stationary household policies and distributions. Its aggregate conditions are:

$$1 = \nu \mathcal{N}(P^*, T^*; \vartheta), \quad (18)$$

$$\bar{H} = Y^* [\mathcal{H}^Y(P^*, T^*; \vartheta) + \mathcal{H}^O(P^*, T^*; \vartheta)], \quad (19)$$

$$2Y^*T^* = \tau^P P^* \bar{H}. \quad (20)$$

The old-state distribution is stationary: $G^* = \Gamma(P^*, T^*; \vartheta)$.

The replacement condition determines whether a positive stationary population is feasible; housing clearing and the balanced rebate jointly characterize its price, transfer, and scale. Because $P_t = P_{t-1}$, equation (4) gives $\ell^* = 0$ in every steady state. Equation (18) also implies that every positive steady state has the same average fertility, $\mathcal{N} = 1/\nu$. A permanent preference change can alter the stationary price, transfer, population scale, and household policies. Conditional on convergence to a positive steady state, demographic stationarity nevertheless requires average fertility to equal $1/\nu$.

Consider two fertility preferences, ϑ^0 and ϑ^1 , for which the steady-state conditions admit positive equilibria $\mathcal{E}^0$ and $\mathcal{E}^1$. At date zero the preference changes permanently:

$$\vartheta_t = \begin{cases} \vartheta^0, & t < 0, \\ \vartheta^1, & t \geq 0. \end{cases} \quad (21)$$

The shock changes current choices immediately and future cohort sizes through equation (16).

Definition 2 (Transition equilibrium). Given the initial steady state $\mathcal{E}^0$, a perfect-foresight transition equilibrium is a sequence of prices, transfers, household policies, old-household distributions, and cohort masses such that all old and young households solve the problems above; the old distribution is generated by the preceding young choices; housing clears and the government budget balances at every date; equation (3) prices the housing asset; equation (16) advances the cohorts; the policy parameters (τ^P, τ^g) remain fixed; $P_{-1} = P^0$, $Y_0 = O_0 = Y^0$, $G_0 = G^0$, and the date-zero old enter with the state inherited from $\mathcal{E}^0$; and the sequence converges to $\mathcal{E}^1$.

The date-zero cohort masses, old states, and tax basis P_{-1} are inherited; P_0 , T_0 , and household choices are equilibrium objects at impact. Capital gains arise only on dates with $P_t > P_{t-1}$ and vanish at both stationary endpoints. The definition imposes convergence but no monotonicity or sign on the transition path.

Intergenerational housing allocation

Consider an old incumbent whose retention and estate-composition constraints are slack with $r_t > \ell_t$, and a young owner whose down-payment constraint binds while the owner-size limit and next-period old constraints are slack. Their marginal values of current housing services are:

$$MV_{it}^Y = r_t + q\ell_{t+1} + \zeta_{it}^{O,F}, \quad MV_{jt}^O = r_t - \ell_t. \quad (22)$$

The market component cancels when a unit is transferred between them.

Definition 3 (Local compensated housing test). At a positive steady state, an allocation passes the local compensated housing test if no infinitesimal transfer of title and occupancy from an old incumbent to a current young owner, together with balanced goods transfers between the two households, can leave both weakly better off and one strictly better off. The comparison holds prices, fertility, cohort masses, tenure labels, and constraint active sets fixed. The current young household's lifetime utility includes its own old age; no unborn household enters the comparison.

Proposition 1 (Steady-state intergenerational reallocation). *Suppose an institution can convey a marginal unit of existing housing from an old incumbent j to a financing-constrained young owner i , finance the transaction within the two-household coalition, and incur real cost K_{ij} per unit. At either positive steady state, the goods-equivalent surplus is:*

$$\mathcal{S}_{ij}^* = \zeta_i^{O,F} - K_{ij}. \quad (23)$$

The steady-state allocation fails the local compensated housing test whenever $\mathcal{S}_{ij}^ > 0$.*

Proof. At a steady state, $\ell_t = \ell_{t+1} = 0$. Equation (22) therefore values an additional unit at $r^* + \zeta_i^{O,F}$ for the young owner and at r^* for the old incumbent. Convey dh units of title and occupancy from j to i , relaxing the buyer's down-payment constraint with a pledgeable bridge $L = (1 - \phi)P^*dh$ at closing and repayment RL next period. The

advance raises the young household's Lagrangian by $(\lambda_i + \mu_i)L$, while the repayment lowers it by $\beta V_{a,i,t+1}^O RL$. The saving Euler equation $\lambda_i = \beta RV_{a,i,t+1}^O$ gives

$$(\lambda_i + \mu_i)L - \beta V_{a,i,t+1}^O RL = \mu_i L = \lambda_i \zeta_i^{O,F} dh.$$

The unit returns to the competitive market at $t + 1$, so later housing allocations are unchanged. The old incumbent sells the same marginal unit. At an interior retention choice, the extra sale proceeds and the change in estate resources compensate her marginal loss of housing services. No realization tax is due at a constant price. Paying the real implementation cost therefore leaves $(\zeta_i^{O,F} - K_{ij})dh$ goods. If this amount is positive, it can be divided so that both households are weakly better off and one is strictly better off. $\square$

Corollary 1 (Transition valuation gap). At any date along the transition, the difference between the two households' marginal valuations is:

$$MV_{it}^Y - MV_{jt}^O = \zeta_{it}^{O,F} + q\ell_{t+1} + \ell_t. \quad (24)$$

Capital gains therefore widen the private allocation gap when the incumbent realizes appreciation at t or the buyer anticipates appreciation at $t + 1$; the gap reduces to $\zeta_i^{O,F}$ at either stationary endpoint.

The corollary does not rank whole transition paths: doing so would require a rule for distributing tax rebates across dates and cohorts.

The reallocation result and the fertility response are distinct. Let $A = 1 + \gamma + \omega_B$ and $k = \beta A$. With interior saving and old choices, substituting the old allocation and the saving Euler equation into either young problem gives the lifetime resource identity:

$$(1 + k)c + \chi n + r_t^m h = \mathcal{W}_t, \quad \mathcal{W}_t = y_i + b_i + T_t + qT_{t+1}. \quad (25)$$

Suppose an access constraint fixes the household's housing at $\hat{h}$ while fertility remains interior. Holding $\mathcal{W}_t$ and prices fixed, implicitly differentiating the fertility condition yields:

$$\frac{dn}{d\hat{h}} = \frac{\frac{\alpha\kappa}{(\hat{h} - \kappa n)^2} - \frac{(1+k)\chi r_t^m}{(\mathcal{W}_t - r_t^m \hat{h} - \chi n)^2}}{\frac{\vartheta_t}{n^2} + \frac{(1+k)\chi^2}{(\mathcal{W}_t - r_t^m \hat{h} - \chi n)^2} + \frac{\alpha\kappa^2}{(\hat{h} - \kappa n)^2}}. \quad (26)$$

The denominator is positive. The first term in the numerator is the value of additional family space; the second is the consumption cost of occupying more housing. Equation (12) implies $[\mathcal{W}_t - r_t^m \hat{h} - \chi n]/[\hat{h} - \kappa n] = (1+k)(r_t^m + \zeta_t^m)/\alpha$. For $r_t^m > 0$, the exact sign condition is:

$$\frac{dn}{d\hat{h}} \geq 0 \iff \chi \leq \frac{(1+k)\kappa(r_t^m + \zeta_t^m)^2}{\alpha r_t^m}. \quad (27)$$

Since $\zeta_t^m \geq 0$, a simpler sufficient condition is:

$$\chi \leq \frac{(1+k)\kappa r_t^m}{\alpha}. \quad (28)$$

Under this condition, relaxing access and permitting the household to purchase more housing at its market user cost raises its fertility. Separately, Proposition 1 implies that a marginal housing transfer passes the local compensated test when $K_{ij} < \zeta_i^{O,F}$. The two conditions can hold simultaneously, but they are distinct results. General-equilibrium prices, rebates, and cohort sizes then propagate that response through equations (16)–(17).

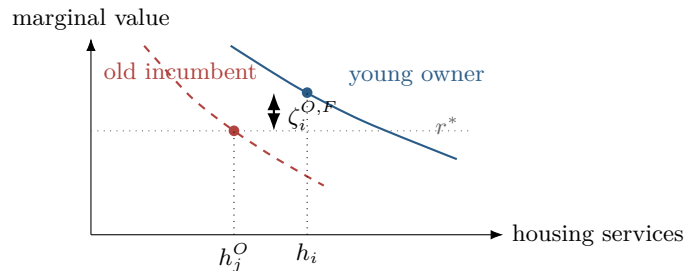


Figure 1: At a positive steady state, the financing wedge raises the constrained young owner's marginal value above the unconstrained incumbent's common user cost. The gap is gross of real implementation cost.